



Maxflow-Mincut Theorem & Ford-Fulkerson

PPT PRESENTED BY

NAME : Srinivas Rao Tammireddy

ROLL NO : 257Y1D5805

YEAR : I Year II Semester

ACADEMIC YEAR : 2025 - 2026

Flow Networks & Constraints

A Flow Network is a directed graph $G = (V, E)$ where each edge has a positive capacity $c(u, v) > 0$. We model the flow of material from a source s to a sink t under two constraints.

Core Mathematical Constraints

A flow f on G must satisfy two properties for all $u, v \in V$:

1. Capacity Constraint: The flow along any edge cannot exceed its capacity:

$$0 \leq f(u, v) \leq c(u, v).$$

2. Flow Conservation: For all nodes except source s and sink t , the total incoming flow must equal the total outgoing flow:

$$\sum f(w, u) = \sum f(u, w). \text{ (No material is stored or lost).}$$

Residual Networks & Augmentation

- **Residual Capacity c_f :** The remaining capacity on an edge:

$$c_f(u, v) = c(u, v) - f(u, v) \text{ (forward capacity),}$$

$$c_f(v, u) = f(u, v) \text{ (unused flow that can be pushed back).}$$

- **Residual Graph G_f :** A graph containing edges with positive residual capacity.

- **Augmenting Path:** A simple path from s to t in the residual graph G_f . Allows us to push additional flow through the network.

The Ford-Fulkerson Method

Ford-Fulkerson iteratively increases flow along augmenting paths until none remain:

Ford-Fulkerson Method Steps

- 1. Initialize Flow:** Set $f(u, v) = 0$ for all edges.
- 2. Find Augmenting Path:** Search G_f for a path P from s to t .
- 3. Find Bottleneck Capacity ($c_f(P)$):** Locate the minimum residual capacity in path P :
$$c_f(P) = \min \{ c_f(u, v) \text{ for } (u, v) \text{ in } P \}.$$
- 4. Augment Flow:** Push $c_f(P)$ along the path. For each edge (u, v) in P , set $f(u, v) += c_f(P)$.
- 5. Repeat:** Reconstruct G_f and repeat until no path from s to t exists.

Edmonds-Karp Optimization

- **Path Search Selection:** Ford-Fulkerson does not define how to search for paths. Using DFS can lead to slow execution times if capacity values are large.
- **Edmonds-Karp Improvement:** Uses Breadth-First Search (BFS) to select the shortest augmenting path (minimum number of edges) in each iteration.
- **Time Complexity:** Edmonds-Karp is guaranteed to run in $O(V \cdot E^2)$ time, completely independent of the edge capacities.

The Maxflow-Mincut Theorem

The Maxflow-Mincut theorem proves the correctness of the Ford-Fulkerson method:

Cuts in Flow Networks

- **Defining a Cut (S, T):** A partition of V into two disjoint sets S and T such that source $s \in S$ and sink $t \in T$.
- **Capacity of a Cut $c(S, T)$:** The sum of capacities of all edges crossing the cut from S to T :
$$c(S, T) = \sum c(u, v) \text{ for } u \text{ in } S, v \text{ in } T.$$
- **Flow Across a Cut $f(S, T)$:** Net flow is the sum of flows from S to T minus the return flows from T to S . For any flow, $f(S, T)$ is equal to the total value of the flow $|f|$.

The Theorem Statement & Equivalence

The Theorem (L.R. Ford & D.R. Fulkerson, 1956):

The maximum value of a flow in a network is equal to the minimum capacity of a cut that separates the source from the sink.

Equivalence Conditions:

1. f is a maximum flow in G .
2. The residual network G_f contains no augmenting paths.
3. $|f| = c(S, T)$ for some cut (S, T) .

Applications of Maxflow

Maxflow-Mincut models many scheduling and resource allocation problems in CS:

Maximum Bipartite Matching

Matches candidates to jobs, or tasks to processors. Model as a flow network by connecting source s to candidates, candidates to jobs, and jobs to sink t with capacity 1. Max flow yields the maximum matching size.

Data Routing in Networks

Determines the maximum data transmission rate between server routers over channels with capacity limits, routing traffic without congestion.

Image Segmentation in CV

Divides an image into foreground and background. Pixels are nodes; edges are color similarities. The min-cut partition separates the foreground from the background optimally.

Airline Crew Scheduling

Schedules flight crews to flight paths while respecting constraints (rest hours, locations). Modeled as a circulation flow network with demands.

THANK YOU

Department of Computer Science and Engineering
Marri Laxman Reddy Institute of Technology and Management